

Solution to the Daily Limit

Based on the problem provided in the file dailyintegral-limit-limit-1x1-2026-08-15.jpg, we are tasked with evaluating the following limit:

$$L = \lim_{n \rightarrow \infty} \frac{1}{n^2} \int_n^{2n} x (\log_n(x))^{\frac{1}{n}} dx$$

1. Analyzing the Integrand

First, we rewrite the logarithmic term using the change of base formula:

$$\log_n(x) = \frac{\ln(x)}{\ln(n)}$$

The integration bounds are from $x = n$ to $x = 2n$. Within this interval, the natural logarithm function is strictly increasing, so we can bound $\ln(x)$:

$$\ln(n) \leq \ln(x) \leq \ln(2n) = \ln(n) + \ln(2)$$

Dividing by $\ln(n)$ (which is strictly positive for $n > 1$), we obtain bounds for our base:

$$1 \leq \frac{\ln(x)}{\ln(n)} \leq \frac{\ln(n) + \ln(2)}{\ln(n)} = 1 + \frac{\ln(2)}{\ln(n)}$$

Raising this inequality to the power of $\frac{1}{n}$ gives us the bounds for the term in our integrand:

$$\begin{aligned} 1^{\frac{1}{n}} &\leq (\log_n(x))^{\frac{1}{n}} \leq \left(1 + \frac{\ln(2)}{\ln(n)}\right)^{\frac{1}{n}} \\ 1 &\leq (\log_n(x))^{\frac{1}{n}} \leq \left(1 + \frac{\ln(2)}{\ln(n)}\right)^{\frac{1}{n}} \end{aligned}$$

2. Setting up the Squeeze Theorem

Since $x > 0$ on the interval $[n, 2n]$, we can multiply the inequality by x without changing the direction of the signs:

$$x \leq x (\log_n(x))^{\frac{1}{n}} \leq x \left(1 + \frac{\ln(2)}{\ln(n)}\right)^{\frac{1}{n}}$$

Next, we integrate all parts of the inequality from n to $2n$, and multiply by $\frac{1}{n^2}$:

$$\frac{1}{n^2} \int_n^{2n} x dx \leq \frac{1}{n^2} \int_n^{2n} x (\log_n(x))^{\frac{1}{n}} dx \leq \frac{1}{n^2} \int_n^{2n} x \left(1 + \frac{\ln(2)}{\ln(n)}\right)^{\frac{1}{n}} dx$$

Notice that the expression $\left(1 + \frac{\ln(2)}{\ln(n)}\right)^{\frac{1}{n}}$ is independent of x and can be factored out of the rightmost integral:

$$\frac{1}{n^2} \int_n^{2n} x dx \leq L_n \leq \left(1 + \frac{\ln(2)}{\ln(n)}\right)^{\frac{1}{n}} \frac{1}{n^2} \int_n^{2n} x dx$$

where L_n is the sequence whose limit we want to evaluate.

3. Evaluating the Bounds

We compute the basic definite integral present on both sides:

$$\int_n^{2n} x \, dx = \left[\frac{x^2}{2} \right]_n^{2n} = \frac{(2n)^2}{2} - \frac{n^2}{2} = \frac{4n^2 - n^2}{2} = \frac{3n^2}{2}$$

Substituting this result back into our inequality yields:

$$\frac{1}{n^2} \left(\frac{3n^2}{2} \right) \leq L_n \leq \left(1 + \frac{\ln(2)}{\ln(n)} \right)^{\frac{1}{n}} \frac{1}{n^2} \left(\frac{3n^2}{2} \right)$$

$$\frac{3}{2} \leq L_n \leq \frac{3}{2} \left(1 + \frac{\ln(2)}{\ln(n)} \right)^{\frac{1}{n}}$$

Now, we evaluate the limit as $n \rightarrow \infty$ for the upper bound term. Let $y = \left(1 + \frac{\ln(2)}{\ln(n)} \right)^{\frac{1}{n}}$. Taking the natural logarithm gives:

$$\ln(y) = \frac{1}{n} \ln \left(1 + \frac{\ln(2)}{\ln(n)} \right)$$

As $n \rightarrow \infty$, the fraction $\frac{\ln(2)}{\ln(n)} \rightarrow 0$, which means $\ln \left(1 + \frac{\ln(2)}{\ln(n)} \right) \rightarrow \ln(1) = 0$. Thus, the product $\ln(y) \rightarrow 0 \cdot 0 = 0$. Because $\ln(y) \rightarrow 0$, it follows that $y \rightarrow e^0 = 1$.

Therefore, the limit of the entire upper bound is:

$$\lim_{n \rightarrow \infty} \frac{3}{2} \left(1 + \frac{\ln(2)}{\ln(n)} \right)^{\frac{1}{n}} = \frac{3}{2} \cdot 1 = \frac{3}{2}$$

Conclusion

Because both the lower bound and the upper bound approach $\frac{3}{2}$ as $n \rightarrow \infty$, we can apply the Squeeze Theorem. The final evaluated limit is:

$$\lim_{n \rightarrow \infty} \frac{1}{n^2} \int_n^{2n} x (\log_n(x))^{\frac{1}{n}} \, dx = \frac{3}{2}$$